

Ansh Agrawal

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Education

Manipal University Jaipur, B.Tech in Computer Science

August 2023 – June 2027

- CGPA: 7.89/10.0 (3rd year) | Coursework: DSA, OOP, DBMS, OS, Computer Architecture, Automata Theory

DAV Public School Sec-49, Gurugram

CBSE

- 12th Grade: 85.2% (2023) | 10th Grade: 95.1% (2021)

Experience

Apollo Tyres Ltd.

May 2026 – Present

Intern – IT & Digital

Gurgaon, India

- Engineered a **RAG pipeline** using **AWS Bedrock Knowledge Base** and **S3** to vectorize graph database schema instructions, enabling semantic retrieval that drives accurate **openCypher query generation** over **Amazon Neptune** property graphs.
- Architected a **hybrid query-routing system** combining RAG-enhanced prompts with dynamic query generation, allowing the agent to autonomously formulate and execute precise graph traversals across complex multi-hop entity relationships.
- Deployed production AI agent using **Strands Agents SDK** on **AWS AgentCore Runtime** with cross-session memory, containerized via **Finch** for Linux ARM64 and served over a live HTTPS endpoint.

Projects

Study Buddy – Voice-Based Agentic Learning Assistant

[GitHub](#)

- **Awarded 3rd place at Assesli Hackathon**; built a 4-phase agentic system (Analyze, Teach, Assess, Recommend) with voice interaction, **Supabase Webhook**-triggered backend, auth, and persistent chat storage.
- Tools: Next.js, Gemini AI, Gemini TTS, Web Speech API, Supabase, LangGraph, Ngrok

FloatChat AI – Natural Language Query System for Oceanographic Data

[GitHub](#)

- **SIH problem statement; cleared internal round.** Built hybrid-routing AI system achieving **95% query success** – simple queries to SQL generation, complex spatial queries to MCP tool orchestration for ARGO float data.
- Tools: FastAPI, PostgreSQL/PostGIS, Groq LLM, ChromaDB, Streamlit, Supabase, LangChain, Plotly

Smart Inventory – AI-Powered Inventory Management Platform

[GitHub](#) | [Live](#)

- **Deployed production SaaS** for SMBs with OCR bill scanning (Tesseract.js + Gemini Vision), atomic recipe-based deduction via **PostgreSQL RPC**, and **3 autonomous AI agents** (reorder, shrinkage, forecasting) with human-in-the-loop approval.
- **BYOK encryption (AES-256-GCM)** with ledger-based stock model via **PostgreSQL triggers**; **Vercel Cron** for scheduled agent execution.
- Tools: Next.js 14, TypeScript, Supabase, Groq LLM, Gemini 2.0 Flash, Tesseract.js, Vercel Cron

Technologies

Languages: C, C++, Java, Python, Kotlin, SQL, JavaScript, TypeScript, HTML/CSS

Technologies: Next.js, Node.js, FastAPI, Flask, Supabase, PostgreSQL, PostGIS, MongoDB, Firebase, TensorFlow, OpenCV, LangChain, LangGraph, ChromaDB, Faiss, Groq, Gemini, Streamlit, Plotly, Flutter, Vercel, Linux, AWS (Bedrock, AgentCore, Lambda, Neptune, S3)

Awards and Achievements

1st Position – The Hackathon: Issued by Manipal University Jaipur · Apr 2025

3rd Position – Agentic Chatbot System: Issued by Assesli · Aug 2025

4th Position – CODESTORM Hackathon: Issued by BITS Goa · Feb 2025

4th Position – ML Hackathon: Issued by IIT Kanpur TechKriti'25 · Mar 2025

Top 5 – Beat the Market Challenge: Issued by IIT Kanpur TechKriti'25 · Mar 2025

Top 10 – Product Challenge: Issued by IIT Kanpur TechKriti'25 · Mar 2025